

MILTON ADINA SHISIA

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SUMMARY

Full-stack software engineer and Computer Science student with a prior B.S. in Epidemiology & Biostatistics. Designs, builds, and ships production web and mobile applications across TypeScript/React/Next.js, Node.js, Java/Spring Boot, and Flutter/Dart - REST APIs, relational data modeling, and strong testing discipline (unit, integration, end-to-end) on automated CI/CD.

SKILLS

Languages:	TypeScript, JavaScript (ES6+), Java, C++, Dart, Python, SQL, Rust (WASM)
Frameworks & APIs:	React, Next.js (App Router), Vue 3, Node.js, NestJS, Express, Fastify, tRPC, Spring Boot, Flutter, Tailwind CSS, REST APIs, serverless/edge functions
Databases & Cloud:	PostgreSQL (Supabase), MySQL, SQLite, Firebase, Pinecone, Neo4j, Cloudflare Workers, Vercel
Testing:	Playwright, Vitest, Jest, Supertest, JUnit 5, Mocha, flutter_test
DevOps & Practices:	Docker, Git/GitHub, GitHub Actions (CI/CD), Agile/Scrum, Maven, Prisma, Drizzle, Sequelize
Security:	OWASP Top 10, threat modeling, vulnerability assessment, multi-tenant Row-Level Security (RLS), Role-Based Access Control (RBAC), SAST/DAST, Semgrep, gitleaks, OWASP ZAP, Software Bill of Materials (CycloneDX SBOM), JWT, bcrypt

CERTIFICATIONS

CompTIA Security+ - In progress (exam planned January 2027).

CLIENT ENGAGEMENTS

Client work: architecture, decisions and dated test evidence at miltonadina.github.io and the linked portfolio showcase.

BrightPath - Multi-Tenant School Management Platform *Client Engagement · Full-Stack & Security Engineer*
TypeScript, React, NestJS, Flutter, Supabase/PostgreSQL, Deno, Docker · Vitest, Playwright · Apr 2025–Present

- Architected a multi-tenant school-management SaaS with 12+ application roles; May 2026 schema audit: **403 tables, 82 migrations and 1,063 RLS policies**. Current NestJS API applies server-side tenant predicates to privileged queries; payment idempotency binds retries to request fingerprints.
- Built tenant-isolation and authorization tests; May 2026 inventory: **2,228 files / 19,639 test declarations**. Built **69 Deno functions** for payments (M-Pesa, Airtel, MTN, Stripe), provider-specific callbacks, and GDPR export/erasure; **260 tests passed** in a selected 19-file run. Reconciled payments and invoices atomically.
- Hardened the platform with a COPPA-focused AppSec workflow: consent-flow testing, Semgrep rules, gitleaks secret scanning, scheduled OWASP ZAP baselines, and dependency CVE auditing in CI/CD.

Lumière - Bilingual Next.js Agency Platform *Client Engagement · Full-Stack Developer*
TypeScript, Next.js 16, Prisma, Supabase (PostgreSQL), Tailwind CSS, Stripe · Dec 2025–Feb 2026

- Built a bilingual (EN/SW) Next.js platform - 40 App Router pages and 25 API routes with React Server Components - with 17 Prisma models and a CMS-style admin so non-technical staff manage services, portfolio, team, and testimonials.
- Implemented Stripe checkout, three rate-limited APIs, input validation, and Supabase Auth with server-side admin API checks.

Flourish - Privacy & Consent-Engineered Family Platform *Client Engagement · Full-Stack & Compliance Engineer*
TypeScript, Turborepo/pnpm monorepo, Fastify, tRPC, Drizzle ORM, PostgreSQL (Supabase), Clerk

- Engineered a TypeScript monorepo with 25 app/package workspaces; May 2026 evidence records **1,278 passing tests** and 25 architecture decisions. Mapped privacy/consent controls to HIPAA, COPPA, CCPA, GDPR, and Washington’s MHMDA.
- Implemented AES-256-GCM field encryption and append-only audit logs (application-role UPDATE/DELETE revocation, immutability test); CI checks include Semgrep, gitleaks, CycloneDX SBOM, and dependency CVE audits.

INDEPENDENT PRODUCT

Light Routines - Cross-Platform Flutter App & Native iOS/Android Bridges *Independent Product · Founder & Sole Engineer*
Flutter/Dart, Kotlin (~740 LOC), Swift (~475 LOC), SQLite, BLE, Firebase · flutter_test · May 2026 counts · Dec 2025–Present

- Built a cross-platform Flutter app in a 5-package monorepo with SQLite persistence and typed MethodChannel/EventChannel interfaces. Implemented device-owner authentication and prototyped Kotlin Foreground Service / Swift CoreBluetooth bridges; the Firebase beta uses a separate Dart session controller.
- Wrote **354 passing tests** (May 2026; 0 analyzer issues). Shipped a Firebase Auth + Firestore closed beta with real testers from June 29, 2026, and an Android internal release.

OPEN SOURCE & COLLABORATION

DevOps - Memory & Runtime Layer for AI Coding Agents

Open-Source Project · Systems Engineer

TypeScript, Fastify, PostgreSQL, npm · MIT-licensed repository · 2026–Present

- Built DevOps and its integrated runtime in an MIT-licensed repository: CLI planning/verification workflows, scoped PostgreSQL memory, and a local provider proxy. Shipped the Claude Code adapter; added loopback Host/Origin checks and optional scoped API-key authentication to the proxy.

Graph Engineering (2026–Present) - Extended a collaborative fork with Dart indexing, content-bound plan approvals, and opt-in reviewed native workers. Changes merged in my fork; upstream PRs #1–2 open.

Hive - Windows quickstart documentation #5668 merged; four open PRs documenting 18 integrations.

COURSEWORK & LABS

Applied AppSec Lab - Authored a STRIDE threat model + data-flow diagram; ran OWASP ZAP against Juice Shop and documented mitigations for CSP, CORS, and cross-origin isolation. Artifacts published in the portfolio showcase.

Doctor Who Knowledge API - Node.js, Express, Sequelize, Supabase, OpenAI; 3-engineer team, 16-model schema. Built responsive UI and OpenAI Q&A from schema/sample rows; JWT routes and 3 passing Jest/Supertest auth tests (mocked DB).

E-Commerce REST API (eBay-style) - Java, Spring Boot, Maven, PostgreSQL, Docker. User/product REST endpoints with bid/order models and repositories; layered architecture, containerized database, and a passing JUnit application-context test.

Internet Applications Portfolio - Vue 3 SPA + vanilla-JS frontends over Express/MySQL in an npm-workspaces monorepo; bcrypt password hashing, list permissions, escaped user content, and origin-allowlisted CORS.

Point-of-Sale (POS) & Inventory System - Java/Swing retail simulation - item management, tax calculation, payment handling, and cashier sessions with CSV persistence and robust parsing/error handling.

Car Dealership Management System - Java Swing + JPA/Hibernate + MySQL with MVC/DAO separation; a JUnit DAO test against an in-memory H2 database.

Additional: Invoice Management System (Java, JUnit); Alien Attack Game (C++, SFML); Dice Roller Pro (Java, SQLite); Data Structures Engine (Java). All public at github.com/MILTONADINA

EXPERIENCE

Freelance Software Engineer (Contract) - Independent Contractor, Remote

2024 – Present

- Deliver production web and mobile applications for private clients end-to-end - architecture, implementation, testing, and deployment - across TypeScript/React/Next.js, Node.js, and Flutter/Dart (detailed client work above).
- Ship with production rigor: multi-tenant Row-Level Security and RBAC, OWASP-aligned hardening, automated SAST/DAST scanning (Semgrep, gitleaks, OWASP ZAP), and full unit/integration/E2E test suites in CI/CD.

USAID HealthIT - Data Analyst Intern, Nairobi, Kenya

Jun 2022 – Nov 2022

- Performed data cleaning and statistical analysis on 5,000+ monthly health-record submissions across 47 county facilities; aligned datasets with Kenya Ministry of Health standards, contributing to a Ministry-recognized 20% report-efficiency improvement.

Freelance Data Analyst & Research Assistant - Various Contracts, Kenya

2020 – 2022

- Designed and deployed mobile data-collection tools (ODK, Kobo, CommCare) for education, health, and industrial research; managed cross-functional teams and delivered analysis reports guiding policy and community programs.

Kisumu County Hospital - Data Entry Specialist Intern, Kenya

Jun 2019 – Nov 2019

- Managed patient data entry into Kenya's EMR system at 98% accuracy, strengthening record integrity; coordinated client enrollment and follow-ups, contributing to an 80% improvement in treatment compliance.

EDUCATION

Oklahoma Christian University

Expected April 2027

B.S., Computer Science (Cybersecurity) - GPA 3.48 · Honor Roll · Ultimate Frisbee Scholarship

Relevant coursework: Network Security, Operating Systems, Computer Organization & Architecture, Software Engineering I–IV, Data Structures & Algorithms, Database Systems, Internet Applications, Cloud Architecture & Security, Artificial Intelligence, Human-Computer Interaction.

Masinde Muliro University of Science and Technology, Kenya

2019

B.S., Epidemiology & Biostatistics - Second-Class Honors

ACTIVITIES & LEADERSHIP

- CS Team - Cyber Contest** - annual collegiate cybersecurity competition, Oklahoma Christian University.
- Ultimate Frisbee Coach**, International School of Kenya (ISK), Mar–May 2023 - mentored student-athletes and organized training programs in a high-standard international environment.
- Ultimate Frisbee Founder & Coach**, MMUST, 2018–2023 - founded and coached a 40+ member team; organized inter-university tournaments.